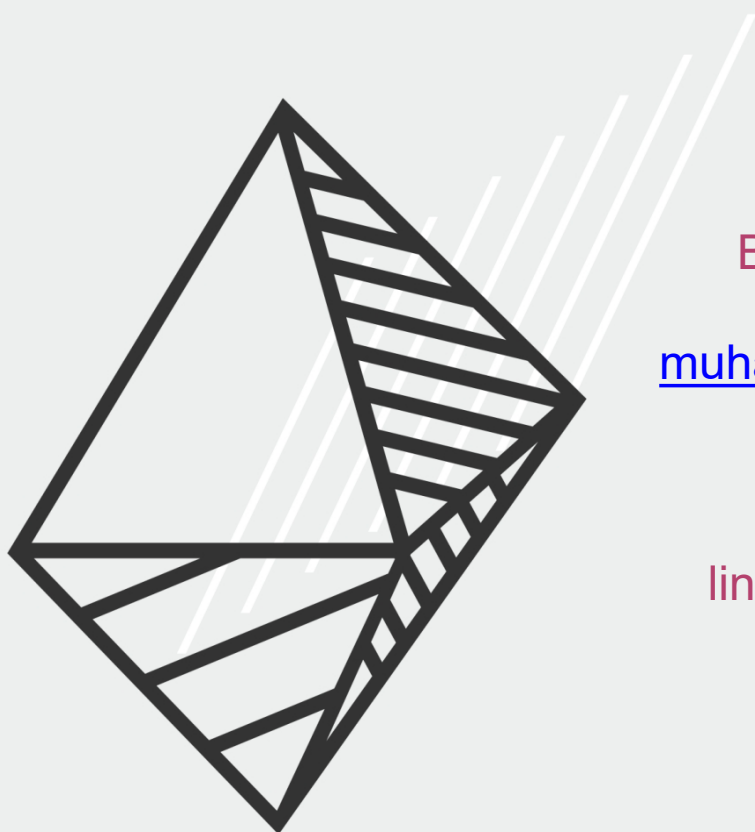


LeafScan

A Plant Disease Detection System
Using Random Forest Classifier

Technical Report



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INTRODUCTION

Agriculture plays a vital role in food production, but plant diseases can significantly reduce crop yield and quality. Farmers often face difficulties in identifying diseases at an early stage, leading to economic losses and excessive use of pesticides. The motivation behind this project is to develop an intelligent and accessible system that can automatically detect plant diseases from leaf images using Artificial Intelligence and Machine Learning. Early disease detection helps farmers take timely preventive measures, improve crop health, and increase agricultural productivity. This system aims to support sustainable farming practices by providing fast, accurate, and cost-effective disease diagnosis.

The Plant Disease Detection System is based on the concept of Machine Learning for automated plant health analysis. Since collecting and labeling a large number of real plant disease images can be time-consuming and expensive, this project utilizes a synthetic dataset generated to represent different plant disease conditions and healthy plants. Various image features such as color, texture, and shape characteristics are extracted from leaf images and used for training machine learning models.

Two ensemble learning algorithms, Random Forest Classifier and Gradient Boosting Classifier, are employed for disease classification. Random Forest improves prediction accuracy by combining the outputs of multiple decision trees, while Gradient Boosting builds a sequence of models that progressively reduce classification errors. The performance of both classifiers is evaluated and compared to identify the most effective model for detecting plant diseases. The developed system provides a fast, reliable, and cost-effective solution for assisting farmers in early disease diagnosis and crop management.

PROBLEM STATEMENT

Plant diseases are one of the major causes of reduced agricultural productivity and crop losses worldwide. Traditional disease identification methods rely heavily on manual inspection by experts, which can be time-consuming, costly, and inaccessible to many farmers. In addition, the lack of large, labeled datasets makes the development of accurate disease detection systems challenging. Therefore, there is a need for an automated and intelligent system that can identify plant diseases quickly and accurately using machine learning techniques. This project aims to address this problem by utilizing synthetic data and machine learning classifiers to detect plant diseases from leaf characteristics.

DESIGN / WAYS & MEANS

The proposed Plant Disease Detection System is designed as a machine learning-based application that analyzes plant leaf features and predicts disease categories. The system uses a synthetic dataset representing healthy and diseased plant conditions. Data preprocessing techniques are applied to prepare the dataset for training. Relevant features are extracted and fed into two machine learning models: Random Forest Classifier and Gradient Boosting Classifier. The trained models are evaluated and compared based on their classification

performance. A user-friendly interface allows users to upload plant images and receive disease predictions along with confidence scores.

BACKGROUND / THEORY

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without explicit programming. In agriculture, machine learning has been widely applied for crop monitoring, disease detection, yield prediction, and resource management.

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree is trained on a random subset of the data, and the final prediction is determined through majority voting.

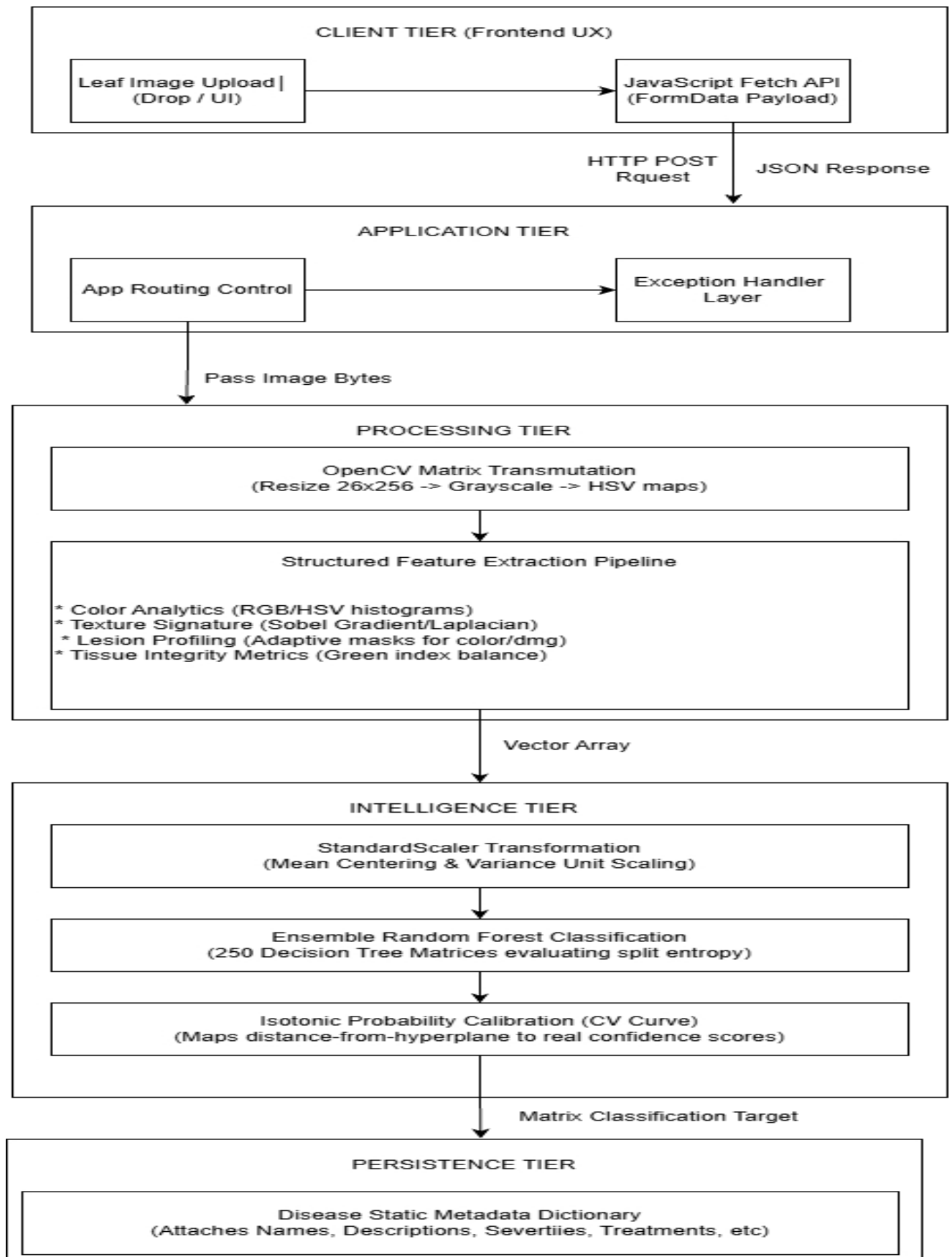
Gradient Boosting is another ensemble learning technique that builds a sequence of decision trees, where each new tree corrects the errors made by previous trees. This approach often achieves high predictive performance by optimizing classification accuracy.

Synthetic data refers to artificially generated data that mimics real-world observations. It is commonly used when collecting large amounts of real data is difficult, expensive, or time-consuming. In this project, synthetic data is used to train and evaluate machine learning models for plant disease detection.

PROCEDURE / METHODOLOGY

1. Generate or collect a synthetic dataset representing healthy and diseased plant leaves.
2. Perform data preprocessing, including cleaning, normalization, and feature preparation.
3. Split the dataset into training and testing sets.
4. Train a Random Forest Classifier using the training data.
5. Train a Gradient Boosting Classifier using the same training dataset.
6. Evaluate both models using performance metrics such as Accuracy, Precision, Recall, and F1-Score.
7. Compare the results of the two classifiers to determine the better-performing model.
8. Integrate the selected model into a web-based application.
9. Allow users to upload plant leaf images and receive disease predictions.
10. Display prediction results and confidence levels through the user interface.

BLOCK DIAGRAM



TEST CASES

Tomato Rotten Leaf

ORIGINAL



Analyze Plant

Clear

LESION ANNOTATIONS



DIAGNOSIS

**Powdery Mildew**
Confidence: 82%

SEVERITY: LOW-MODERATE

Fungal disease causing white powdery coating on leaf surfaces, stunting growth and reducing photosynthesis.

RECOMMENDED TREATMENTS

PREVENTION

→ Apply potassium bicarbonate or neem oil spray

→ Use sulfur-based fungicide as preventative

→ Improve air circulation around plants

→ Remove heavily infected leaves

Plant in sunny, well-ventilated locations; avoid overhead irrigation.

TOP PREDICTION PROBABILITIES

Powdery Mildew	82.0%
Healthy	13.3%
Late Blight	4.7%
Early Blight	0.0%
Leaf Mold	0.0%

Random Plant Leaf

ORIGINAL



LESION ANNOTATIONS



Analyze Plant

Clear

DIAGNOSIS

**Late Blight**
Confidence: 83.8%

SEVERITY: HIGH

Phytophthora infestans causes water-soaked lesions that turn dark brown/black, spreading rapidly in cool, wet conditions.

RECOMMENDED TREATMENTS

PREVENTION

→ Apply systemic fungicide (mancozeb or metalaxyl) immediately

→ Remove all infected plant material and dispose off-site

→ Do not compost infected material

→ Increase spacing between plants for better airflow

Use certified disease-free seeds; plant resistant varieties.

TOP PREDICTION PROBABILITIES

Late Blight	83.8%
Powdery Mildew	14.3%
Leaf Mold	1.2%
Healthy	0.7%
Early Blight	0.0%

MODEL EVALUATION

REAL DATASET PERFORMANCE EVALUATION				
Overall Validation Accuracy: 94.56%				
Detailed Per-Class Performance:				
	precision	recall	f1-score	support
Healthy	0.97	0.98	0.98	644
Early Blight	0.93	0.91	0.92	400
Late Blight	0.90	0.93	0.92	582
Leaf Mold	0.96	0.97	0.97	191
Bacterial Spot	0.96	0.95	0.95	625
Septoria Leaf Spot	0.96	0.93	0.95	354
accuracy			0.95	2796
macro avg	0.95	0.95	0.95	2796
weighted avg	0.95	0.95	0.95	2796

ANALYSIS

The analysis phase focuses on evaluating the effectiveness of the machine learning models in detecting plant diseases. Both Random Forest and Gradient Boosting classifiers are tested using unseen data to measure their generalization capability. Performance metrics such as accuracy, precision, recall, and F1-score are used to compare the models. The results help determine which classifier provides better prediction performance and reliability. The analysis also examines the impact of synthetic data on model training and identifies potential improvements for future work, such as incorporating real-world datasets and deep learning techniques to enhance detection accuracy.